INCH-POUND

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DETAIL SPECIFICATION

INDICATORS AND TRANSMITTERS: LIQUID, TEMPERATURE AND PRESSURE

Inactive for new design after 22 January 1992.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers liquid quantity, temperature, and pressure indicators and transmitters for 24 volt (nominal) direct current (DC) electrical systems. The transmitters are of the electrical resistance type.

1.2 Classification. Indicators and transmitters are of the following units, types, and classes as specified (see 6.2).

1.2.1 Unit.

Unit 1 - Indicator
Unit 2 - Transmitter

1.2.2 Type.

Type I - Liquid quantity
Type II - Temperature
Type III - Pressure

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil/.

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1.2.3 Class.

Class 1 - Float type
Class 2 - Submerged tube type

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

L-P-380	- Plastic Molding Material Methacrylate.

FEDERAL STANDARDS

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-595	- Colors Used in Government Procurement (Color Chips 37875 and 37038).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (For Indicating Instrument).
MIL-DTL-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-DTL-64159	- Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant.

(See Supplement 1 for list of specification sheets.)

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product.

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D 3935	- Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material.

(Copies of this document are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

IPC - ASSOCIATION CONNECTING ELECTRONICS INDUSTRIES

J-STD-006	- Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.

(Copies of this document are available online at http://www.ipc.org/ or from IPC - Association Connecting Electronic Industries, 3000 Lakeside Drive, 309 S, Bannockburn, IL 60015.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the applicable specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.4 Materials. Materials shall be as specified herein, in applicable specifications, and specification sheets. Materials not specifically designated shall be suitable for use in indicators and transmitters operating over ranges specified without any change in physical or dimensional properties that would result in operation of the units falling outside of the limits specified herein.

3.4.1 Radioactivity. Indicators furnished under this specification shall not contain radioactive material.

3.4.2 Solder. Solder used in making electrical connections shall conform J-STD-006 and shall be rosin fluxed.

3.5 Construction. Construction of the indicators and transmitters shall conform to the form and dimensions in applicable specification sheet, as specified (see 6.2).

3.5.1 Threaded surfaces. All threaded surfaces shall be in the inch system and shall conform to FED-STD-H28.

3.5.2 Indicator.

3.5.2.1 Dial.

3.5.2.1.1 Dial markings. Unless otherwise specified (see 6.2), numerals, letters, and graduation marks shall be clearly defined white on a black background. The dial face shall be marked as shown in the applicable specification sheet. Dial markings such as numerals, letters and graduation marks not covered by MIL-STD-130 shall have a minimum height of 1/8 inch. Numerals, letters, and graduation marks shall conform to FED-STD-595 color number 37875. The dial face shall conform to FED-STD-595 color number 37038.

3.5.2.1.2 Fastening. The dial face shall be secured to the body of the indicator in such a position that the markings will be legible when illuminated through the case windows from an outside source. The dial shall not become loose or turn when subjected to vibration or shock (see 4.4.10.4 and 4.4.11).

3.5.2.1.3 Pointer. The pointer shall be attached to the indicator movement and shall be dampened to reduce oscillations from vibration. The pointer shall not touch the dial face. The shape of the pointer shall not interfere with reading any point in the range of the dial. A minimum of one-half of the visible portion of the pointer shall conform to FED-STD-595 color number 37875.

3.5.2.1.4 Cover window. The cover shall be treated on both sides with an electrostatic solution unless the window material has been tested for static charge generating characteristics and has been demonstrated to be non-susceptible to a static charge that would deflect the pointer when the window is rubbed with a cloth containing not less than 95 percent fleece or pulled wool.

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The window shall be secured to the base so that moisture is excluded from the instrument case. The thickness of the window shall be 1/8 inch minimum. The window shall have an Izod strength of not less than 0.3 foot-pounds (ftlb) and shall retain transparency with age. The window material shall conform to MIL-W-80, or if molded, the material shall conform to type I, class 3 of L-P-380 or ASTM D 3935.

3.5.2.2 Case. The case shall be constructed of steel suitable for the purpose or plastic molding material conforming to ASTM D 3935.

3.5.3 Transmitter.

3.5.3.1 Mounting.

3.5.3.1.1 Type I. Type I transmitters shall be provided with a flange to permit attachment to fuel tanks. The diameter of the flange, bolt holes, and bolt hole spacing shall be as specified in the applicable specification sheet.

3.5.3.1.2 Types II and III. Type II and III transmitters shall be provided with a male standard pipe thread connection of the size specified in the applicable specification sheet.

3.6 Performance. Indicators and transmitters covered by this specification shall meet the performance requirements at all rated levels of temperature, pressure, and electrical input as specified herein and in applicable specification sheets.

3.6.1 Calibration.

3.6.1.1 Indicator. Unless otherwise specified in the applicable specification sheet, when tested in accordance with 4.4.2, the centerline of the pointer shall indicate the centerline of the appropriate dial graduation ±3 degrees. When current is off, the pointer shall return to the extreme left position.

3.6.1.2 Transmitter. When tested in accordance with 4.4.3, the transmitter resistance measurements shall conform to the applicable specification sheet.

3.6.2 Voltage variation, indicator. When tested in accordance with 4.4.4, the indicator shall conform to 3.6.1.1 with an additional allowable tolerance of +2/-4 degrees.

3.6.3 Overload capacity, transmitter.

3.6.3.1 Type II. When tested in accordance with 4.4.5.1, the type II transmitter shall conform to 3.6.1.2 and shall show no evidence of mechanical or electrical damage after exposure for one minute to a temperature of 350 ˚F (177 ˚C).

3.6.3.2 Type III. When tested in accordance with 4.4.5.2, the type III transmitter shall withstand a pressure of 200 percent of the maximum pressure specified in the applicable specification sheet and shall show no evidence of mechanical or electrical damage. The transmitter shall conform to 3.6.1.2 after this test.

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3.6.4 Attitude error, indicator. When tested in accordance with 4.4.6, the indicator shall show no evidence of damage and shall conform to 3.6.1.1 with an additional allowable tolerance of ±2 degrees.

3.6.5 Reverse connection capability, indicator. When tested in accordance with 4.4.7, the indicator shall show no evidence of damage and shall conform to the requirements of 3.6.1.1 with an additional allowable tolerance of ±1 degree.

3.6.6 Dielectric strength. When tested in accordance with 4.4.8, the insulation of indicators and transmitters with metal cases only shall show no evidence of cracking or charring.

3.6.7 Endurance. The indicators and transmitters shall met the requirements of 3.6.1 with the following allowable tolerances after being tested in accordance with 4.4.9.

a. Type I, II, and III indicators shall have an additional allowable tolerance of ±2 degrees.
b. Type I transmitters shall have an additional allowable tolerance of +2.0 ohms at empty and ±1 ohm at full.
c. Type II transmitters shall have an additional allowable tolerance of ±1 percent of the specified resistance at each calibration point.
d. Type III transmitters shall have an additional allowable tolerance of +0.5 ohm at zero indicator pressure and ±1.0 ohm at other checkpoints specified in the applicable specification sheets.

3.6.8 Vibration resistance. Type I transmitters shall be tested in accordance with 4.4.10.1. Type II transmitters shall be tested in accordance with 4.4.10.2. Type III transmitters shall be tested in accordance with 4.4.10.3. Indicators shall be tested in accordance with 4.4.10.4. While indicators and transmitters are being vibrated, instruments shall be monitored to determine lapse of operation. Lapse of operation shall constitute test failure. Subsequent to the test, measurements shall be taken to determine whether the indicators and transmitters meet the calibration requirements of 3.6.1 with the added allowable tolerances specified in 3.6.7. The indicators and transmitters shall show no evidence of mechanical or electrical damage throughout the test.

3.6.9 Shock resistance. Type I, II, and III indicators and type II and III transmitters shall be tested in accordance with 4.4.11. After testing, indicators and transmitters shall show no evidence of mechanical or electrical damage and shall meet the requirements of 3.6.1 with added tolerance margins as follows:

a. Type I, II, and III indicators shall have an additional allowable tolerance of ±2 degrees.
b. Type II transmitters shall have an additional allowable tolerance of ±1 percent of the specified resistance at each calibration point.
c. Type III transmitters shall have an additional allowable tolerance of +2 ohms at zero or -1 ohm at the other checkpoints specified in the applicable specification sheets.

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3.6.10 Temperature influence.

3.6.10.1 Indicator. When tested in accordance with 4.4.12.1, the indicator shall operate throughout an ambient air temperature range of -65 ˚F to 160 ˚F (-54 ˚C to 71 ˚C). Indicators shall conform to 3.6.1 with an additional allowable tolerance of ±5 degrees.

3.6.10.2 Transmitter. When tested in accordance with 4.4.12.2, the type I transmitter shall operate throughout an ambient temperature range of -65 ˚F to 160 ˚F (-54 ˚C to 71 ˚C). The type III transmitter shall operate throughout an ambient temperature range of -65 ˚F to 260 ˚F (-54 ˚C to 127 ˚C). Transmitters shall conform to 3.6.1 with the additional allowable tolerances of +1 ohm at empty or zero or ±1 ohm at the other checkpoints specified in the applicable specification sheets after testing.

3.6.11 Waterproofness.

3.6.11.1 Type I, II, and III indicators and type II transmitters. When tested in accordance with 4.4.13.1.1, type I, II, and III indicators and type II transmitters shall show no evidence of leakage and shall be mechanically and electrically operable during and subsequent to submersion.

3.6.11.2 Type I transmitter. When tested in accordance with 4.4.13.2.2, type I transmitter components including floats, tubes, and covers shall show no evidence of leakage.

3.6.11.3 Type III transmitter. When tested in accordance with 4.4.13.1.2, type III transmitters shall show no evidence of leakage and shall subsequently meet the requirements of 3.6.10.2 for all calibration points on the applicable specification sheet.

3.6.12 Corrosion resistance. When tested in accordance with 4.4.14, indicators and transmitters shall conform to 3.6.1 and shall show no evidence of corrosion affecting operation.

3.6.13 Fungus resistance. When tested in accordance with 4.4.15, indicators and transmitters shall conform to 3.6.1 and shall show no evidence of fungus growth affecting operation.

3.6.14 Transient voltages, indicator. When tested in accordance with 4.4.16 and 4.4.17, indicators shall show no evidence of damage or loss of calibration.

3.7 Finish. The indicator bezel shall be lusterless camouflage forest green in accordance with MIL-DTL-53039 or MIL-DTL-64159.

3.8 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall assure that the product is free of rust, burrs, and scratches. All electrical connections shall be mechanically secured prior to being soldered or brazed to prevent them from coming loose due to vibration or shock. All solder cleaning agents and flux shall be non-corrosive. All traces of flux shall be chemically neutralized and removed after soldering.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When first article inspection is required (see 3.2), the sample shall consist of four specimens of each type and class of unit to be tested. The inspection shall consist of the tests in the order listed in table I for indicators and table II for transmitters.

TABLE I. Order of first article testing of indicators.

Specimen no.	Requirement paragraph	Test	Test paragraph
1	3.6.1.1	Calibration	4.4.2
	3.6.2	Voltage variation	4.4.4
	3.6.10.1	Temperature influence	4.4.12.1
	3.6.7	Endurance	4.4.9.1
2	3.6.1.1	Calibration	4.4.2
	3.6.4	Attitude error	4.4.6
	3.6.9	Shock resistance	4.4.11
	3.6.11.1	Waterproofness	4.4.13.1.1
	3.6.14	Positive transient voltage	4.4.16
	3.6.14	Negative transient voltage	4.4.17
3	3.6.6	Dielectric strength	4.4.8
	3.6.1.1	Calibration	4.4.2
	3.6.8	Vibration resistance	4.4.10.4
	3.6.12	Corrosion resistance	4.4.14
	3.6.5	Reverse connection capability	4.4.7
4	3.6.1.1	Calibration	4.4.2
	3.6.13	Fungus resistance	4.4.15
	3.6.1.1	Calibration	4.4.2

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TABLE II. Order of first article testing of transmitters.

Specimen no.	Requirement paragraph	Test	Test paragraph
1	3.6.1.2	Calibration	4.4.3
	3.6.10.2	Temperature influence	4.4.12.2
	3.6.7	Endurance	4.4.9.2
2	3.6.6	Dielectric strength	4.4.8
	3.6.1.2	Calibration	4.4.3
	3.6.8	Vibration resistance:	
		Type I	4.4.10.1
		Type II	4.4.10.2
		Type III	4.4.10.3
	3.6.12	Corrosion resistance	4.4.14
3	3.6.1.2	Calibration	4.4.3
	3.6.9	Shock resistance	4.4.11
	3.6.11	Waterproofness:	
		Type I	4.4.13.2.2
		Type II	4.4.13.1.1
		Type III	4.4.13.1.2
	3.6.3	Overload capacity	4.4.5
4	3.6.1.2	Calibration	4.4.3
	3.6.13	Fungus resistance	4.4.15
	3.6.1.2	Calibration	4.4.3

4.3 Conformance inspection. Conformance inspection shall consist of the examinations of 4.3.2 and the tests of 4.3.4, as applicable.

4.3.1 Sampling for units.

4.3.1.1 Inspection lot. A lot shall consist of all indicators or transmitters of one type and class, from one identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for the examinations of 4.3.2 shall be selected in accordance with level I of MIL-STD-1916.

4.3.1.3 Sampling for tests. From each lot that has passed the examinations of 4.3.2, a representative sample for testing shall be selected in accordance with level I of MIL-STD-1916.

4.3.2 Examination. Each indicator or transmitter selected in accordance with 4.3.1.2 shall be examined for the defects listed in table III for conformance to this specification and to the applicable specification sheet.

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TABLE III. Classification of defects.

Characteristic	Method of inspection	Category	
		Major	Minor
Dimensions affecting interchangeability not within tolerance (see 3.5)	Gage	101	
Dimensions not affecting interchangeability not within tolerance (see 3.5)	Gage		201
Defective painting (see 3.5.2.1.1, 3.5.2.1.3, and 3.7)	Visual		202
Improper marking (see 3.8)	Visual		203
Faulty workmanship (see 3.9)	Visual		204

4.3.3 Classification of conformance tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.3.4).
b. Control tests (see 4.3.5).

4.3.4 Acceptance tests. Samples selected in accordance with 4.3.1.3 shall be tested as specified in 4.4.2 or 4.4.3, 4.4.4, and 4.4.13.2.1 or 4.4.13.2.2, as applicable.

4.3.5 Control tests.

4.3.5.1 Sampling for control tests. Control test samples shall be selected at the rate of three of each 500 items produced of a particular unit, type, and class except that not less than three or more than six items shall be selected in any 30-day period. The samples shall be selected from a lot that has passed the examinations of 4.3.2 and shall be subjected to the separate tests of 4.3.5.2.

4.3.5.2 Procedure for control tests. The three items selected for control tests shall undergo the tests listed in table IV for indicators and table V for transmitters.

TABLE IV. Control tests for indicators.

Item no.	Requirement paragraph	Test	Test paragraph
1	3.6.1.1	Calibration	4.4.2
	3.6.8	Vibration resistance	4.4.10.4
2	3.6.1.1	Calibration	4.4.2
	3.6.11.1	Waterproofness	4.4.13.1.1
	3.6.14	Positive transient voltage	4.4.16
	3.6.14	Negative transient voltage	4.4.17
3	3.6.1.1	Calibration	4.4.2
	3.6.11.1	Waterproofness	4.4.13.1.1
	3.6.10.1	Temperature influence	4.4.12.1

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TABLE V. Control tests for transmitters.

Item no.	Requirement paragraph	Test	Test paragraph
1	3.6.1.2	Calibration	4.4.3
	3.6.8	Vibration resistance:	
		Type I	4.4.10.1
		Type II	4.4.10.2
		Type III	4.4.10.3
2	3.6.1.2	Calibration	4.4.3
	3.6.11	Waterproofness:	
		Type I	4.4.13.2.2
		Type II	4.4.13.1.1
		Type III	4.4.13.1.2
3	3.6.1.2	Calibration	4.4.3
	3.6.11	Waterproofness:	
		Type I	4.4.13.2.2
		Type II	4.4.13.1.1
		Type III	4.4.13.1.2
	3.6.10.2	Temperature influence	4.4.12.2

4.4 Tests.

4.4.1 Test conditions. Tests shall be conducted in laboratory free of dust, drafts, fumes, and products of combustion. Ambient room temperature shall be maintained at 77 ˚F ± 15 ˚F (25 ˚C ± 8 ˚C). Test specimens shall be stabilized for one hour at this temperature before being tested. Readings shall be taken on indicators and from type II transmitters only after voltage has been applied for a minimum of five minutes. Indicators and type I and type III transmitters may be lightly tapped while readings are being taken.

4.4.1.1 Calibration of test equipment. Test equipment resolution shall permit measurement of 10 percent of the product tolerance. Instruments such as voltmeters, ammeters, and thermometers shall be of the laboratory type and shall be accurate to within ±0.50 percent. Protractors shall be accurate to within ±0.25 degrees and shall be free from error due to parallax. Test equipment calibration records made at intervals sufficient to establish the required accuracy shall be made available to the government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the government.

4.4.2 Indicator calibration.

4.4.2.1 Apparatus. The apparatus for calibration shall include a test chamber with an air temperature range of -65 ˚F to 160 ˚F, (-54 ˚C to 71 ˚C), a protractor, voltmeter, ammeter, a decade resistance box, and a source of 28 volts direct current (VDC) ± 0.5 VDC.

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4.4.2.2 Procedure. To determine conformance to 3.6.1.1, the indicator shall be tested in accordance with the following procedure:

a. The indicator shall be mounted in a vertical position.
b. The temperature shall be held at 77 ˚F ± 15 ˚F (25 ˚C ± 8 ˚C).
c. The unit terminals shall be connected to the voltage source and resistance applied as specified in the applicable specification sheet.
d. The protractor shall be used to take dial pointer readings at each calibration point.

4.4.3 Transmitter calibration.

4.4.3.1 Type I transmitter.

4.4.3.1.1 Apparatus. The apparatus for calibration shall include a fuel tank equipped with a fixture for attaching the transmitter, a wheatstone bridge or ohmmeter, and test liquid with a specific gravity of 0.721 ± 0.0036.

4.4.3.1.2 Procedure. To determine conformance to 3.6.1.2, the transmitter shall be tested in accordance with the following procedure. Calibration shall be done on a descending scale.

a. The transmitter shall be attached to the fixture installed in the fuel tank.
b. The fuel tank shall be filled with liquid until the float (class 1) or the tube (class 2) is at the full position.
c. The resistance of the transmitter shall be measured with a wheatstone bridge or ohmmeter.
d. The fuel tank shall be emptied and the resistance of the transmitter measured again. Float (class 1) transmitters may be tested at the "full" and "empty" positions without being installed in the fuel tank.

4.4.3.2 Type II transmitter.

4.4.3.2.1 Apparatus. The apparatus for calibration shall include an oil tank equipped with the means of controlling the oil temperature and to recirculating the oil, a milliammeter, a voltmeter, a supply of engine lubricating oil, and a 28 VDC ± 0.5 VDC electrical power source.

4.4.3.2.2 Procedure. To determine conformance to 3.6.1.2, the transmitter shall be tested in accordance with the following procedure. Calibration shall be on an ascending scale.

a. The transmitter shall be immersed in a tank of oil that is at the lowest temperature specified in the applicable specification sheet.
b. The oil shall then be circulated at a minimum rate of 10 feet per minute past the transmitter.
c. The oil will then be heated to each of the temperatures specified in the applicable specification sheet.
d. The internal resistance of the transmitter shall be measured at each temperature.

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4.4.3.3 Type III transmitter.

4.4.3.3.1 Apparatus. The apparatus for calibration shall include a dead weight tester or an oil or air tank with provisions for controlling internal pressure, a precision pressure gage, and a wheatstone bridge or ohmmeter.

4.4.3.3.2 Procedure. To determine conformance to 3.6.1.2, the transmitter shall be tested in accordance with the following procedure. Calibration shall be on an ascending scale starting at the lowest pressure specified in the applicable specification sheet.

a. With the transmitter installed in the pressure tank, the pressure shall be increased to the pressure values specified in the applicable specification sheet. The pressure shall be held at each level for one minute.
b. The internal resistance of the transmitter shall be measured at each pressure level with a wheatstone bridge or an ohmmeter.

4.4.4 Indicator voltage variation. To determine conformance to 3.6.2, the indicator shall be tested in accordance with 4.4.2 except the unit shall be connected first to a 20 VDC and then a 32 VDC source. A reading shall be made at each voltage value.

4.4.5 Transmitter overload capacity.

4.4.5.1 Type II transmitter. To determine conformance to 3.6.3.1, the transmitter shall be tested in accordance with 4.4.3.2 except that it shall be held in a 350 ˚F (177 ˚C) oil bath for one minute prior to the test. The transmitter shall be examined following the test.

4.4.5.2 Type III transmitter. To determine conformance to 3.6.3.2, the transmitter shall be tested in accordance with 4.4.3.3 except that it shall be subjected to an oil or air pressure of 200 percent of the pressure specified for a full scale reading on the applicable indicator. The transmitter shall be examined following the test.

4.4.6 Attitude error, indicator. To determine conformance to 3.6.4, the indicator shall be mounted in a manner that allows the face to be positioned as follows:

a. 45 degrees clockwise from normal (vertical plane).
b. 45 degrees counterclockwise from normal (vertical plane).
c. Face up (horizontal plane).
d. Face down (horizontal plane).

With the exception of the mounting position, the indicator shall be tested in accordance with 4.4.2. Dial pointer readings shall be taken for each of the four dial face positions.

4.4.7 Reverse connection capability, indicator. To determine conformance to 3.6.5, the indicator power source and signal connections shall be reversed for a period of two hours. A 28 VDC ± 0.5 VDC power source shall be applied to the signal terminal and the lowest resistance specified in the applicable specification sheet shall be applied to the power terminal on the indicator.

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4.4.8 Dielectric strength for indicators and transmitters with metal cases only. To determine conformance to 3.6.6, the insulation shall be examined after the following test. This test is to be conducted prior to final assembly for sealed units.

a. The lead within the indicator of transmitter case shall be disconnected from ground or the ground terminal.
b. A potential of 440 V alternating current at a frequency of 60 Hertz shall be applied between the case and the ungrounded circuit or winding for one minute.

4.4.9 Endurance.

4.4.9.1 Indicator. To determine conformance to 3.6.7, the indicator shall be tested in accordance with 4.4.2 after the following cycling period:

a. The indicator shall be electrically connected to the applicable transmitter or equivalent resistance.
b. The indicator shall be cycled to full scale deflection and return for a total of 200,000 cycles. One cycle shall consist of one complete oscillation from extreme left to extreme right and back to extreme left.
c. The cycling rate shall allow full scale deflection of the pointer.

4.4.9.2 Transmitter. To determine conformance to 3.6.7, the transmitter shall be tested in accordance with 4.4.3 after the following cycling period:

a. The transmitter shall be cycled for a total of 100,000 cycles (10,000 cycles for type II transmitters).
b. One cycle shall consist of one full scale deflection and return on an applicable indicator.
c. The cycling rate shall provide full scale deflection of the applicable indictor pointer.

Transmitters may be tested concurrently with their applicable indicators.

4.4.10 Vibration.

4.4.10.1 Type I transmitter. To determine conformance to 3.6.8, the transmitter shall be tested in accordance with the following procedure:

a. The transmitter shall be mounted in a manner to simulate actual use.
b. The transmitter shall be vibrated in the direction of each of the major axes in accordance with MIL-STD-202, Method 201, "Vibration".
c. The transmitter shall be tested in accordance with 4.4.3.1 at the end of each vibration period along each of the three axes and while the transmitter is still being vibrated.

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4.4.10.2 Type II transmitter. To determine conformance to 3.6.8, the transmitter shall be tested in accordance with the following procedure:

a. The transmitter shall be mounted in a test fixture simulating actual mounting in use. Care shall be taken to ensure the mounting is free from resonance over the frequency range.
b. The transmitter shall be connected in a series circuit with a 28 VDC ± 0.5 VDC power source and an applicable indicator.
c. With the circuit energized, the transmitter shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch ± 0.003 inch or 50 g peak, whichever is less. Tolerance for the amplitude required to reach 50 g is ±10 percent. The vibration frequency shall be varied logarithmically over the range of 10 cycles per second (cps) to 3,600 cps. If there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used.
d. The vibration cycle for 10 cps to 3,600 cps and back to 10 cps shall be accomplished in 20 minutes ± 2 minutes. This scanning cycle shall be repeated three times so that the critical frequencies (see 6.4) can be identified, recorded, and checked.

4.4.10.2.1 Critical frequency procedure. If critical frequencies are specified, the following procedure shall be used:

a. After the three scanning cycles, the transmitter shall be vibrated at the critical frequency for two hours.
b. If there is more than one critical frequency, the two-hour period shall be divided equally between the critical frequencies.
c. If there are more than three critical frequencies, the transmitter shall be vibrated for 40 minutes at each of the three most critical frequencies.

4.4.10.2.2 No critical frequency procedure. If no critical frequency is identified, the following procedure shall be used:

a. The transmitter shall be vibrated at 50 g with a frequency cycle of 1,400 cps to 3,400 cps and back to 1,400 cps. Tolerance for the amplitude required to reach 50 g is ±10 percent. The rate of frequency change shall be logarithmic. If there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used.
b. The vibration cycle for 1,400 cps to 3,400 cps and back to 1,400 cps shall be accomplished in 20 minutes ± 2 minutes. This scanning cycle shall be repeated six times for a total test time of two hours.
c. At the end of the two-hour vibration period, the transmitter shall be tested in accordance with 4.4.3.2.

4.4.10.2.3 Total test time. The test shall be performed on each perpendicular axis for a total test time of 9 hours, 3 hours per axis.

4.4.10.3 Type III transmitter. To determine conformance to 3.6.8, the transmitter shall be tested in accordance with MIL-STD-202, Method 204, "Vibration, High Frequency", test condition A. The transmitter shall be rigidly mounted in the normal operating position and connected in series with a 28 VDC ± 0.5 VDC power source and an applicable indicator. After completion of the vibration cycles, the transmitter shall be tested in accordance with 4.4.3.3.

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4.4.10.4 Indicator. To determine conformance to 3.6.8, the indicator shall be mounted vertically and vibrated in each of the three major axes in accordance with MIL-STD-202, Method 201, "Vibration". At the end of each vibration period, the indicator shall be tested in accordance with 4.4.2.

4.4.11 Shock. To determine conformance to 3.6.9, indicators and transmitters shall be tested in accordance with the following procedure:

a. Type I, II, and III indicators and type II and III transmitters shall be mounted in a manner to simulate actual mounting in use.
b. The indicator or transmitter shall be shock tested in accordance with MIL-STD-202, Method 213, "Shock (Specified Pulse)", so that an impact of 100 g is obtained.
c. The test shall be repeated ten times and applied on the vertical axis while the indicator or transmitter is in its normal mounted position.
d. The indicator or transmitter shall then be tested in accordance with 4.4.2 or 4.4.3, as applicable.

4.4.12 Temperature influence.

4.4.12.1 Indicator. To determine conformance with 3.6.10.1, the indicator shall be tested in accordance with 4.4.2 except that the ambient air temperature shall be held at -65 ˚F ± 5 ˚F (-54 ˚C ± 3 ˚C) and subsequently held at 160 ˚F ± 5 ˚F (71 ˚C ± 3 ˚C). The indicator shall be soaked in the test chamber for four hours until its temperature is stabilized at that of the ambient air. Readings shall be made at each temperature.

4.4.12.2 Transmitter. To determine conformance to 3.6.10.2, the transmitter shall be tested in accordance with 4.4.3 except that the ambient temperature and the transmitter temperature shall be stabilized and the resistance of the transmitter measured at the following temperatures:

a. Type I - 160 ˚F ± 5 ˚F (71 ˚C ± 3 ˚C) and -65 ˚F ± 5 ˚F (-54 ˚C ± 3 ˚C).
b. Type III - 260 ˚F ± 5 ˚F (127 ˚C ± 3 ˚C) and -65 ˚F ± 5 ˚F (-54 ˚C ± 3 ˚C).

4.4.13 Waterproofness.

4.4.13.1 First article and control test.

4.4.13.1.1 Type I, II, and III indicators and type II transmitters.

4.4.13.1.1.1 Apparatus. The apparatus shall consist of a chamber equipped with a viewing window through which the indicator or transmitter may be observed while it is under test. The chamber shall be equipped with suitable seals, terminal connections, gauges, and pumps to permit either evacuation or pressurization as required. The minimum size of the chamber shall be 100 percent in excess of the test component and its hardware.

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4.4.13.1.1.2 Saline solution. The salt used shall be sodium chloride containing on a dry basis not more than 0.1 percent sodium iodide and not more than 0.2 percent total impurities. The solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of distilled water or other water containing not more than 200 parts per million of total solids. The solution shall be kept free of sediment by filtration or decantation.

4.4.13.1.1.3 Procedure. To determine conformance to 3.6.11.1, type I, II, and III indicators and type II transmitters shall be tested in accordance with the following procedure:

a. The indicator or transmitter, with its electrical connections, shall be submerged in a container with the uppermost surface a minimum of one inch below the surface of saline solution (see 4.4.13.1.1.2) and shall be installed in the test chamber.
b. The indicator or transmitter shall be observed during the entire period of submersion and shall be operated while submerged for 30 minutes at full rated current and voltage.
c. The chamber shall be evacuated to a pressure of 6 pounds per square inch (psi) below atmospheric in order to apply a minimum of 6 psi in internal pressure to all voids within the indicator or transmitter.
d. During the test period, the indicator or transmitter shall be observed for poor seals as evidenced by bubbles escaping from the interior of the indicator or transmitter. Leakage thus indicated shall be considered noncompliance with 3.6.11.1 and the indicator or transmitter shall be rejected. Bubbles that are the result of entrapped air on the surfaces of the indicator or transmitter shall not be considered a leak. Test results shall be compared with the operating results of 4.4.2 or 4.4.3, as applicable.
e. The chamber shall then be pressurized to 6 psi above atmospheric and the component again operated for 30 minutes. Test results shall be compared with the operating results of 4.4.2 or 4.4.3, as applicable.

4.4.13.1.2 Type III transmitter. To determine conformance with 3.6.11.3, the transmitter shall be tested in accordance with the following procedure:

a. The apparatus specified in 4.4.13.1.1.1 shall be used.
b. The threaded end of the transmitter shall be capped during the test.
c. The transmitter shall be submerged in tap water with the uppermost surface a minimum of one inch below the surface of the water.
d. The transmitter shall be checked for leaks by applying 18 pounds internal pressure for 30 minutes. The internal pressure shall be obtained by applying a vacuum over the water.

4.4.13.2 Acceptance test.

4.4.13.2.1 Type I, II, and III indicators and type II and III transmitters. To determine conformance to 3.6.11.1, type I, II, and III indicators and type II and III transmitters shall be tested in accordance with 4.4.13.1.1 except that tap water shall be used and the internal pressure shall be 6 psi for five minutes. Observation for leaks shall be made during this five-minute period. Bubbles that are the result of entrapped air on the surfaces of the indicator or transmitter shall not be considered a leak.

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4.4.13.2.2 Type I transmitter. To determine conformance to 3.6.11.2, type I transmitter floats, tubes, and covers shall be tested for leakage by immersion in the saline solution of 4.4.13.1.1.2 at a temperature of not less than 200 ˚F (93 ˚C). Covers shall be tested for leakage under a tank pressure of 6 psi when the cover is mounted on the tank. Observation for leaks shall be made during this five-minute period. Bubbles that are the result of entrapped air on the surfaces of the indicator or transmitter shall not be considered a leak.

4.4.14 Corrosion resistance. To determine conformance to 3.6.12, indicators and transmitters shall be tested in accordance with MIL-STD-810, Method 509, "Salt Fog", except that the test duration shall be 200 hours. When the corrosion test follows the vibration test, the additional allowable tolerances of 3.6.7 shall be applicable. Following the test, the indicators and transmitter shall meet the requirements of 3.6.1.1 or 3.6.1.2, as applicable.

4.4.15 Fungus resistance. To determine conformance to 3.6.13, indicators and transmitters shall be tested in accordance with MIL-STD-810, Method 508, "Fungus", except the test period shall be 90 days. Following the test, indicators and transmitters shall meet the requirements of 3.6.1.1 or 3.6.1.2, as applicable.

4.4.16 Positive transient voltage. To determine conformance to 3.6.14, the indicator shall be connected in a test circuit as shown in figure 1. The indicator shall be subjected to five pulses by cycling switch S1 from "B" to "A". The switch shall remain in each position for three minutes. The indicator shall then be tested in accordance with 4.4.2.

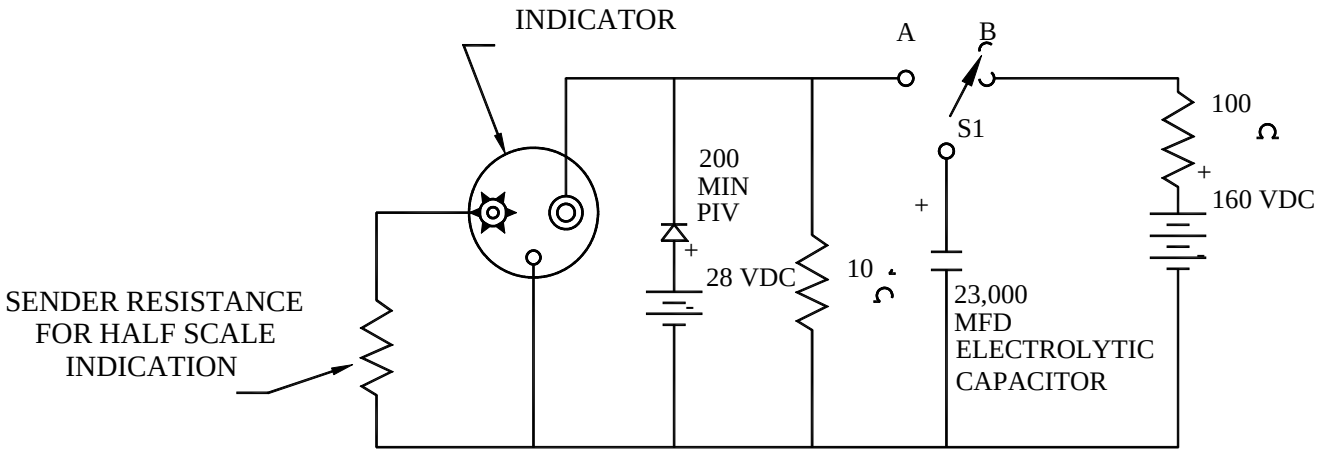


FIGURE 1. Positive transient test circuit.

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4.4.17 Negative transient voltage. To determine conformance to 3.6.14, the indicator shall be connected in a test circuit as shown in figure 2. The indicator shall be subjected to five pulses by cycling switch S2 from "D" to "C". The switch shall remain in each position for one minute. The indicator shall then be tested in accordance with 4.4.2.

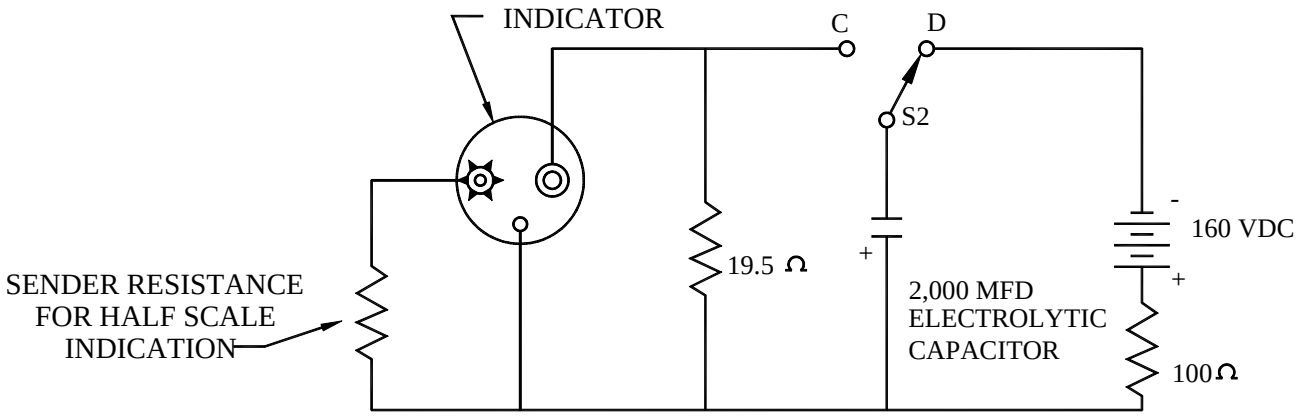


FIGURE 2. Negative transient test circuit.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. An indicator (dashboard unit) and a transmitter (sending unit), when connected by a single conductor cable (not supplied) and properly grounded, comprise a liquid quantity, temperature, or pressure gage. These devices are used on military motor vehicles and other internal combustion applications to inform the operator of the following:

Type I - Level of fuel in supply tank or tanks.
Type II - Engine coolant, transmission coolant, or other liquid coolant temperature.
Type III - Oil pressure in the engine or transmission lubricating system or the air pressure in the airbrake system.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Unit, type, and class (see 1.2).
c. First article inspection, when required (see 3.2).
d. Applicable specification sheet (see 3.5).
e. Dial markings, if different (see 3.5.2.1.1).
f. Packaging (see 5.1).

6.3 First article. When a first article is required (see 3.2), the contracting officer should include specific instructions in all procurement documents regarding arrangements for examination, testing, and approval of the first article.

6.4 Definitions.

6.4.1 Critical frequency. Critical frequency is a resonant frequency developed in the transmitter such that the amplitude exceeds that of the applied vibration. The greater this amplitude, the more highly loaded the transmitter and the more critical the vibration environment.

6.5 Subject term (key word) listing.

dashboard unit
dial face
float type
motor vehicle
sending unit
submerged tube type

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - YD

Preparing Activity:
DLA - GS1

(Project 6620-2007-002)

Review Activities:
Army - CR, CR4
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

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